ASSIGNMENT 1

□ Title :- Configuring Print Server

□ Procedure :-

- At first configure IP address, mask value, default gateway in Both server & client → Then open server and roles (ADDS, DNS, Print & document) → Promote ~~in~~ the server as ~~a~~ AD server → create reverse lookup zone on DNS.
- Server :- Tools → Print management → Expand Print servers → RC on Printers → add printer → Add a ~~TCP/IP or~~ new Printer using an existing port ~~web services~~ ...... → ~~select type of device (TCP/IP device) → give IP address~~ → install a new driver → next → Microsoft → Microsoft XPS class driver → change to share this file → next → Next → Finish → RC on added Printer → Properties → List in the directory.
- windows run → dsa.msc → users → new → user → name (user1), same as ~~to~~ name of logon name → next give password.
- Properties of user → Dial in → Allow access → Member of → add → Administrator → check name → Ok.

• Client : - Add in bwu.com domain → login with administrator@bwu.com → control panel → Hardware devices → Printers → add printer → select → Next → Print a text page → then on server you can see the job.

ASSIGNMENT 2

Title:- Configuring http Server

Procedure:-

At first we have to create a EC2 instance -> click on Launch instances -> give name of the Server -> Select the OS image (windows server 2016) -> Next create keypair and keep the file format ".Pem" -> apply default subnet and VPC -> Then create the security rules (rdp, http, https) -> After that launch the instance and status should be displayed 3/3 check.

Taking remote connection:-

Click on connect after select EC2 -> RDP client -> click on get password & upload private key -> then take remote connection of the server.

IIS installation:-

Open Server Manager -> click add roles & features -> then install web service in the server -> then create a html file under the C:\inetpub\wwwroot -> open tools & click IIS Manager -> click on Server & add website -> give site name (: bwu.com) & choose Path (inetpub) -> select ip address & hostname (www.bwu.com) -> After that you can Access website from browser.

19/01/2026

ASSIGNMENT 7

□ Title:- Configuring Active Directory system-state backup and restoration.

□ Procedure :-

Server1	IP - 192.168.13.1 DNS - 192.168.13.1

bwu.com Role - ADDS, DNS

- At first configure IP address and chage server name -> Then install ADDS, DNS role & ppromote to AD with bwu.com.
- Tools -> Active Directory users & computers -> RC on bwu.com -> New -> organizational unit (sales_OU) -> RC on sales OU -> New -> user -> sal1 -> make one more user sal2.
- Then add one more 20GB hard disk through VM settings. -> after that open server & give command on windows run -> diskmgmt.msc & Online it -> RC on Disk1 RC on disk1 & initialize it -> Then RC on Disk 1 & select new simple volume -> Next...... -> ok.
- Then add roles & features -> Next -> Next -> Next -> (Not found in ~~role~~, you found in features section after roles) -> select windows server backup & install.

ASSIGNMENT 3

Title:- RAID Configuration

Procedure:-

Server

At first create a EC2 ~~with~~ on AWS and 6 (10GB) storage along with primary storage (30GB) → then take RDP of Server → then open Tools → Computer management → Click Disk management → online all the added disk (RC on disk & online) → the initialize select all disk & choose GPT → ok (convert all disk to dynamic disk)

Striped volume (RAID 0):- RC on disk 1 → New striped volume → Next → click Disk 2 & add → Next → Next → Next → Finish → yes (For Dynamic disk conversion.

Mirrored volume (RAID 1):- ~~RC on~~ If you use disk2 for mirrored volume, you need to delete volume. RC on disk2 → then you need to convert dynamic disk all the disk → then RC on disk2 → New mirrored volume → Next → select disk3 & add → Next → Next → Next → finish

Parity volume (RAID 5):- If you use disk3, delete volume. ~~RC on~~ → RC on Disk → New RAID-5 volume → Next → add disk4 & disk5 → Next → Next → Next → finish

Rel 09/02

ASSIGNMENT 4

Title:- Configuring Windows Deployment Service (WDS).

Procedure:-

Client

bwu.com IP - 192.168.13.1

Server DNS - 192.168.13.1

Role - ADDS, DHCP, DNS, WDS

Configure Server:- • At first configure the DC as per the Scenario (change name, IP address, etc) → Then add roles ADDS, DHCP, DNS, WDS → After that Promote the server to DC → then open DNS from tool → and Right click on reverse lookup zone → create new zone ~~as~~ per need → R RC on new zone & add a new pointer record → IP (192.168.13.1) and browse & select server.

• After that complete DHCP config on server manager Dashboard & ~~commit~~ Commit → then click tool & open DHCP → → RC on server.bwu.com and click IPV4 & new ~~st~~ scope → next → scope1 (name) → next → 192.168.13.100 to 192.168.13.200 → Next → Next → next → yes → next → finish.

• After that open VMware server settings and browse 'window 10' ISO (For install on client machine)

• Tools → WDS → expand server → RC on server → configure server → Next → Next → yes → next → ..... → next → finish. respond to all client computers

• → RC on install image → install image → next → browse → this PC → DVD (D) drive → sources → install.wim →

open → next → ~~win~~ select only win pro → next. ····→ RC on boot → add boot image → boot.wim → next → next → finish → then RC on server → all task & start.

Create a new machine on VMware → select ~~tipical~~ typical → I will install the OS later → windows & windows 10 x64 → next → name & select Path → next → finish → then open & start the machine → then you enter in ~~boot option~~ bios → enter setup & change boot order → enter → EFI network (top) ~~using sh~~ (shift (+) +) → enter → commit changes exit → exit the boot maintainance manager → reset the system → after that windows is install.

ASSIGNMENT 5

Title :- Active Directory Sites Configuration.

Procedure :-

KOL-SERVER	MUM-SERVER	DEL-SERVER
Network - VMNET3	Network - VMNET 4	Network - VMNET5
ADDS, DNS	ADDS, DNS	ADDS, DNS
DC - KOL	DC - MUM	DC - DEL
IP - 192.168.13.1/24	IP - 192.168.17.1/24	IP - 192.168.19.1/24
DG - 192.168.13.100	DG - 192.168.17.100	DG - 192.168.19.100
DNS - 192.168.13.1	DNS - 192.168.13.1	DNS - 192.168.13.1

BWU.COM

Router

Role - Remote Access
VMNET 3 - 192.168.13.100/24
" 4 - 192.168.17.200/24
" 5 - 192.168.19.100/24

Router Configuration :- • At first add VMNET3 → ncpa.cpl → Rename kolkata-site → give ip 192.168.13.100 & subnetmask 255.255.255.0 → then VMNET 4 → Rename Mumbai-site (192.168.17.100 & 255.255.255.0) → add VMNET 5 → Rename (del-site (192.168.19.100 & 255.255.255.0)

- After that add role remote access → Next select Routing add → Next ..... → install
- ~~Tools → ADSS → Expand site~~
- Tools → Routing & Remote access → Expand server → RC on server local → configuration enable routing and remote access → Next → custom configuration → Next → LAN Routing → Next → Finish → start server

Kolkata-server :- At first assign IP address & add roles (ADDS, DNS) → then promote bwu.com → Tools → ADSS → expand sites → RC on ~~site → Default-site~~ Default-first-site-name & rename Kol-site → RC on site → New site (Mum-site) → ok → ok → RC on site → New site (del-site) → ok → ok → RC on subnet → New subnet → Prefix (192.168.18.0/29) → select kol-site → RC on subnet → New subnet → (192.168.17.0/29) & select Mumbai → RC on subnet → New subnet → Prefix (192.168.19.0/29) & select del-site.

Mumbai-server :- Add roles (ADDS, DNS) → Promote & select add a domain controller to an existing domain → bwu.com & select → administrator@bwu.com & password → give DSRM Password → Next → Replicate from kolkata-server → Next → Next → install.

Delhi-server :- same as mumbai-server.

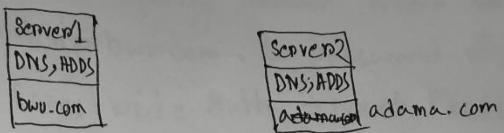
Kolkata-server :- ~~Exp~~ open ADSS → expand del-site → server → Delhi-server → RC on NTDS settings & select Replicate configuration from selected DC & ~~Kol-server~~ kolkata-server & same steps for delhi server.

Red 10/03/26

ASSIGNMENT 6

□ Title :- Configuring trust relationship between forests.

□ Procedure :-



- At first asign corresponding IP address, default gateway, mask value in both server → Then install DNS, ADDS role and promote as a new forest - server1 (bwu.com) & server2 (adama.com) → then create reverse lookup zone & add new pointer ~~record~~ record
- After that you need to add stub zone in server1 & server2.

Server 1 :- Tools → DNS → RC on forward Lookup zone → New zone → Next → select stubzone → Next → Next → zone name (adama.com) → ip or adama.com.

Server2 :- Same as server1 but change zone name (bwu.com) & ip or bwu.com.

- Then need to configure trust relation between forests, you can configure server1 or server2. I select server2 for configuration.

Server 2 :- Tools → Active Directory Domain & Trust → expand the forest → RC on forest → Properties → Trust → New Trust → Next → Root domain Name (bwu.com) → Next

→ choose Trust type (Forest Trust) → Mention the way (Two-way) → Next → both this domain and specified domain → Next → specify domain Name ~~Ad~~ (bwu.com) → Administrator@bwu.com & password ~~of~~ (bwu.com Password) → Forestwide Auth → Next Finish.

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ASSIGNMENT 8

Title :- Create virtual machine using Hyper-V.

Procedure :-

Server
BWU.com
192.168.13.1

Role - ADDS, DNS, Hyper-V
VMNET 3

Requirement :- • (i) 64 bit processor, (ii) Hardware virtualization support (Intel. VT-X/AMD-V), (iii) SLAT
• Search windows features and verify those are unchecked. (Hyper-V, Virtual Machine ~~platform~~ Platform, windows Hypervisor platform, windows sandbox)

Then open vmware setting and Add a Network adapter (VMNET 4) & one harddisk → after that open ~~server~~ windows run (diskmgmt.msc) → open disk management & online disk 1 → intialize disk & make simple volume. → make a folder (virtual-machine) on this disk/drive.

After Active Directory & DNS configuration need → open tools → Hyper-V manage → RC on server → HyperV setting → virtual Hard Disk → select the folder ~~folder~~ & folder ~~(virtual_Machine)~~ in disk 1 → virtual_Machine folder (virtual_Machine) → Ok.
→ select the same folder → virtual switch manager → create virtual

RC on server → virtual switch manager → create virtual switch → External (External_virtual switch) → External network (Intel(R). 82574L Gigabit Network connection 2)

open vmware setting add iso (windows 10) →

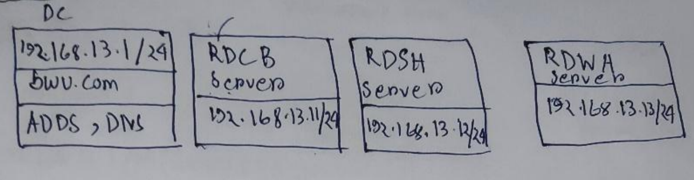
RC on Hyper-V manager -> new -> virtual Machine -> windows 10 -> Generation 1 -> startup memory (2048 MB) -> connection (External-virtual-switch) -> harddisk size (50 GB) -> install an operation system from bootable CD/DVD ROM -> select D drive -> Next ... -> RC on windows 10 -> start

Red 14/03

ASSIGNMENT 9

Title :- Configuring remote desktop services with an RD-session host ~~session~~ server.

Procedure :-



At configure DC , then add the three ~~server~~ server.

DC :- All server RC → add server → Find NOW → ctrl + select all → ok → Manage → add role & features → → next → Next → Remote Desktop service installation → standard Deployment → Next → session-based desktop deployment → next → select RDCB.bwu.com → Next → select → RDWA.bwu.com → Next → select RDSH.bwu.com → Next → ~~select~~ ~~select~~ check (Restart .....) → Deploy → close → click on Remote Desktop services (from Dashboard) → click on RD Session host → create session collection → Name (Enterprise-Apps) → Next → add RDSH → Next → Next → uncheck Enable user profile disk → Next → Create → Enterprise-Apps → publish remote app Programme → add wordpad & more → Publish → create an user on AD → open web-browser & type ~~http://RDWA~~ (http://RDWA.bwu.com/rdweb) → login using (user1 & pass@word1)

Backup :-
• open server → open powershell on Run as administrator → use wbadmin /? for check all command → use Command wbadmin start systemstatebackup -backupTarget:E: → write y for continue → use wbadmin get versions for check backup

open ADUC → view → advance features → RC on sales_OU & select properties → navigate to object → uncheck Protect object from accidental deletion → ok → then RC on sales_ou and delete.

Restoration :-
• windows Run → msconfig → Navigate to Boot → select safe Boot → select Active directory repair → ok → Restart → other user → username → administrator & Password Pass@word1 (DSRM) → windows Run & CMD → use wbadmin get versions & copy version identifier → use wbadmin start systemstaterecovery -version: 03/06/2026-04:49 (version identifier) & y & y → restart login on other user (Previous credential) → windows Run → msconfig → uncheck safeboot → ok → Restart

06/03/26

ASSIGNMENT 10

Title :- Configuring VPN.

Procedure :-

VPN Server	Router	Client
Role - ADDS, DNS, Remote access	Role - Remote Access	
192.168.13.1/24 (VMNET3)	192.168.13.100 (VMNET3)	172.16.0.101/16 (VMNET4)
DG - 192.168.13.100	172.16.0.100 (VMNET4)	

- At first configure ADDS, DNS role on VPN server.
- Then give ip on VMNET3 at first, then add VMNET4 and give ip.
- Then only give ip on client machine.

Router :- Manage -> Add Roles & Features -> Next -> Next -> -> Next -> select Remote Access -> Next..... -> select Routing -> Next.... -> install

- Tools -> Routing & Remote access -> configure & enable routing & remote address -> custom configuration -> Next -> check LAN Routing -> Next -> Finish -> start service

Server :- install remote access same as Router only select DirectAccess & VPN (RAS).

- click on open the Getting started wizard -> Deploy VPN only -> RC configure & enable Routing & remote access -> custom configuration -> check VPN access
- Tools -> ~~Remo~~ Routing & remote access -> RC on server -> Properties -> Navigate to security tab -> check Allow custom IPSEC........ -> Preshared key (12345678)

→ IPv4 tab → Static address pool → add → Start IP (192.168.13.90) & End IP (192.168.13.99) → ok → apply → ok

• Tools → Active Directory user & computer → user → New (user1) → user1 Properties → Dail-in → Network Access Permission → check allow access.

Client :- open control panel → Network & internet → Network & sharing centre → set up a new connection or network → connect to a work place → Next → use my internet connection → I'll set up an internet : ..... → Next → internet address (192.168.13.1) → create

click on internet.

~~start~~ setting → Network & internet → VPN → connection → Name (BWUVPN) → server address (192.168.13.1) → VPN Provider (windows (built-in)) → VPN type (L2TP/IPSec with Pre-shared key

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ASSIGNMENT 11

Title :- Configuring Routing and DHCP Relay agent.

Procedure :-

DC
DHCP Server

192.168.13.1/24

ADDS, DNS, DHCP

DUC-192.168.13.100

Router

LANS segment 1
192.168.13.100/24
172.16.0.100/16
LAN segment 2

~~Remote &~~
~~Routing~~
Role- Remote access

client 1
automatic
IP addressing

Scope-1 (1st Floor)

192.168.13.0/24

Scope-2 (2nd Floor)

172.16.0.0/16

Router :- Tools → Routing & Remote Access → RC on Router & click ~~to~~ Configure and Enable Routing ..... → Next → custom configuration → LAN routing → Finish

DC :- Tools → DHCP → click on IPV4 → New scope → scope1 → start IP (192.168.13.101) & End IP (192.168.13.150) (192.168.13.100) & add → → Next .... → Router IP (192.168.13.100) & add → Next ..... → Finish

Tools → DHCP → click on IPV4 → New scope → scope2 → start IP (172.16.0.101) & End IP (172.16.0.150) → Next .... → Router IP ~~192.168.13~~ (172.16.0.100) & add → Next ..... → Finish

Router :- ~~Routi~~ Tools → Routing & remote access → IPV4 → General → RC new ~~interface~~ routing protocol → RC on DHCP Protocol → DHCP Relay agent

relay agent → Ethernet 1 → ok → OK → RC on DHCP
relay agent ~~PO~~ → Properties → server address →
192.168.13.1 → add

~~client~~
client: · Disable NIC → then enable

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ASSIGNMENT 12

Title :- Configuring network load balancing & failover load balancing.

Procedure :-

DC
server1
bwu.com
192.168.13.1/24
Role- ADDS, DNS

member server — Role-web server
NODE1
192.168.13.11/24
Features- Network load balancing

member server — Role - web server
NODE 2
192.168.13.12/24
Features- Network load balancing

At first configure AD & DNS in server1 → Then add this two Member server with the domain → install network load balancing features & web server in those member server → after that login on member servers (administrator@bwu.com)

* use 'sysprep' in any two system if you using same ovf for those three system.

NODE 1 :- At first create node1.html file on ~~C:\inetp~~ C:\inetpub\wwwroot location → then open tools → IIS manager → NODE1 → Default document → RC and add node1.html.

NODE2 :- Same steps only chage. file name node2.html.

NODE1 :- Tools → Network Load balancing manager → RC on Network Load balancing cluster → New → Host (192.168.13.11) → Next → IPV4 address (192.168.13.99) → select Full internet name (www.bwu.com) → Next → edit → From (80) To (80) → Finish. multicast → Next

RC on www.bwu.com → Add Host to cluster → Host (192.168.13.12) connect → Next → Next → finish.

Server 1 :- Tools → DNS → server1 → bwu.com → add new host → www & IP address (192.168.13.99) → Add host

Rew 14/03

ASSIGNMENT 12
Failover clustering

Server1	Member Server	Member Server
BWU.com	Node1	Node2
192.168.13.1	192.168.13.11/24 (VMNET 3)	192.168.13.12/24 (VMNET3)
Role- ADDS, DNS,	10.0.0.1 (VMNET 4)	10.0.0.2 (VMNET 4)
ISCSI Target server, ISCSI target Storage Provider	Features - Failover Clustering	Features - Failover clustering

* At first configure ADDS, DNS at server1 → install role SICSI Target server & ISCSI target storage provider under the File & storage service → Then open VMware setting and add 60 GB HardDisk → after that open disk manager. online disk, initialize and create new simple ~~vol~~ volume.

* Open Server1 Dashboard → File & storage services → ISCSI → New ISCSI virtual disk → custom path → Browse → Select New volume (E:) & select path → Next → Name (Vdisk1) → Next → Size 30 GB → next → New ISCSI target → next → ~~Access~~ ~~server (Add)~~ → Target name (BWU-SAN) → Next → Access server (add) → enter value of selected type → ~~IP address~~ type (IP address) → value (192.168.13.11) → OK → add → type (IP address) → value (192.168.13.12) → Next .... → create → close

* Create another one using same process for 5 GB storage.

Node1 :- Add a network adapter from vmware setting (VMNET4) & give ip 10.0.0.1.

Node2 :- Same steps as ~~Node~~ Node1, only change IP address 10.0.0.2

* you need to add features Failover clustering in both Node1 & Node2.

Node1 :- Tools → iSCSI → iSCSI intialize → target (192.168.13.1) → connect → done → ok

Node2 :- Same steps as Node1

Node1 :- open disk manager → online, intialize & create simple value in both disk (30GB & 5 GB)

Node2 :- open disk manager and only online disk

Ren 16/03

Node1 :- Tools → Failover cluster manager → create cluster → Next → browse → Find (NODE1, NODE2) → ok → Next → select ~~no~~ → Second option → bwucluster (cluster Name) → ~~192.168~~ Address 192.168.13.100 (192.168.13.1/29) → Next → Next → Finish → click on ~~cluster~~ RC on cluster bwucluster.bwu.com → Disk → RC → add to cluster shared disk1 (available storage) → add to cluster shared ~~drive~~ then click volume → ~~create folder on~~ file server.
Role → configuration role → next → file server.

⊞ Backup :-
• open server → open powershell on Run as administrator → use wbadmin ø /? for check all command → use command wbadmin ø start ø systemstatebackup ø ø -backupTarget:E: → write y for continue → • use wbadmin ø get ø versions for check backup

⊞ open ADUC → view → advance features → RC on sales_OU & select properties → navigate to object → uncheck Protect object from accidental deletion → ok → then RC on sales_ou and delete.

⊞ Restoration :-
• windows Run → msconfig → Navigate to Boot → select safe Boot → select Active directory repair → OK → Restart → other user → username administrator & Password Pass@word1 (DSRM) → windows Run & CMD → use wbadmin ø get ø versions & copy version identifier → ~~use~~ use wbadmin ø ø start ø systemstaterecovery ø -version: 03/06/2026-04:49 ~~&~~ y (version identifier) & y & y → restart login using on other user (Previous credential) → windows Run → msconfig → uncheck safeboot → OK → Restart

Deh 06/03/26